

School of Physics and Astronomy

Understanding the growth and electronic properties of rare earth Kagome metals

PHS3350 Research Project

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Abstract

Kagome lattices show potential for use in next-generation electronics due to their unique electronic properties. One of the most promising Kagome metals is thin-film $TbMn_6Sn_6$. Analysis of bulk rare-earth Kagome metal $TbMn_6Sn_6$ has already confirmed it exhibits a flat band, and a Dirac cone in its band structure. Also, a band gap is expected to open in the material when $TbMn_6Sn_6$ is atomically thin. The consequence of this is that the thin-film variation of the material is expected to have infinitely heavy electrons, which have applications in quantum computing, and also potential for electronic currents to flow without energy loss – integral for fast and low-energy computing.

Synthesis and investigation into the thin-film variation of this material is essential to determine how these features evolve with reduced thickness, as the band gap is expected to depend on film thickness. In this study, thin films of $TbMn_6Sn_6$ were grown using Molecular Beam Epitaxy and the chemical and electronic structure characterised via X-ray and Angle-Resolved Photoemission Spectroscopy. X-ray Photoemission Spectroscopy (XPS) revealed prominent Mn, Sn and Tb core levels consistent with $TbMn_6Sn_6$, however a slight shoulder in the Sn was observed. The origin of this shoulder was investigated using detailed peak fitting and considering the Sn bonding environments. Angle-Resolved Photoemission Spectroscopy (ARPES) measurements revealed preliminary evidence of the predicted flat bands and Dirac cone features. These results demonstrate successful synthesis of high-quality near-2D films and highlight the influence of thickness on the chemistry. Further high-resolution ARPES and depth-dependent XPS studies are proposed to confirm the full band structure and chemical environment.

Acknowledgements

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1 Introduction

For decades, progress in digital electronics has been driven by miniaturisation: making transistors and circuits smaller, faster, and more energy-efficient [1] . From centimetre-scale devices in 1945 to nanometre-scale architectures today, hardware scaling has enabled the explosive growth of modern computing. However, at these dimensions, quantum tunnelling, heat dissipation, and device-to-device variability impose hard physical limits. At the same time, the demands of artificial intelligence, large-scale data storage, and high-speed computation continue to rise, creating pressure for entirely new electronic materials.

One promising direction is 2D Kagome metals. Their name originates from a traditional Japanese basket-weaving pattern of interlaced triangles and hexagons. Replicating this geometry in an atomic lattice produces exotic electronic behaviour, and since their first successful synthesis in 2018 [2], Kagome metals have become a rapidly expanding research field.

Kagome band structures host a dispersionless flat band and a Dirac cone. In the flat band, electrons have effectively infinite mass and are spatially localised, leading to strong electron–electron interactions and many-body quantum states with potential applications in quantum information [3]. Near the Dirac point, electrons behave as massless Dirac fermions, enabling high mobility and low-dissipation charge transport. In atomically thin Kagome systems, a band gap opens between Dirac cones, giving rise to topologically protected edge states that are separated from the bulk states. Electrons can hence flow in these edge states without scattering, resulting in dissipationless currents.

This project focuses on the rare-earth Kagome metal $TbMn_6Sn_6$, specifically in thin-film form. Although bulk $TbMn_6Sn_6$ has already been synthesised and confirmed to host both a flat band and a Dirac cone, its thin-film variant must be grown and characterised to determine whether a band gap forms and whether dissipationless transport can be achieved.

2 Theory

2.1 Rare-Earth Kagome Lattice

2.1.1 Lattice Structure

Rare-Earth Kagome Metals are a family of Kagome metals and are identified by the chemical formula RMn_6Sn_6 where R = rare earth. The rare earth metals are comprised of the lanthanides, Sc, and Y and are distinguished by their partially filled 4f electron shells (shielded inside 5s and 5p orbitals), which create robust ferromagnetic order in the Kagome lattice whereby the spins of atoms align out-of-plane, leading to Dirac cones, flat bands, and topological transport. In this project, we study R = Tb and the lattice structure for the unit cell is in figure 1. The Mn atoms form the Kagome geometry,

arranged as corner-sharing triangles in a 2D layer. These Mn Kagome sheets are stacked along the c-axis and separated by Sn layers and Tb-Sn blocks. The structure is hexagonal and layered, consisting of repeating Mn Kagome sheets, intervening Sn coordination, and Tb-based spacer layers. There are three distinct bonding environments: Tb-Sn, Sn-Sn and Mn-Sn outlined in figure 1. Both intra-layer and inter-layer bonds exist between atoms, forming the 3D structure.

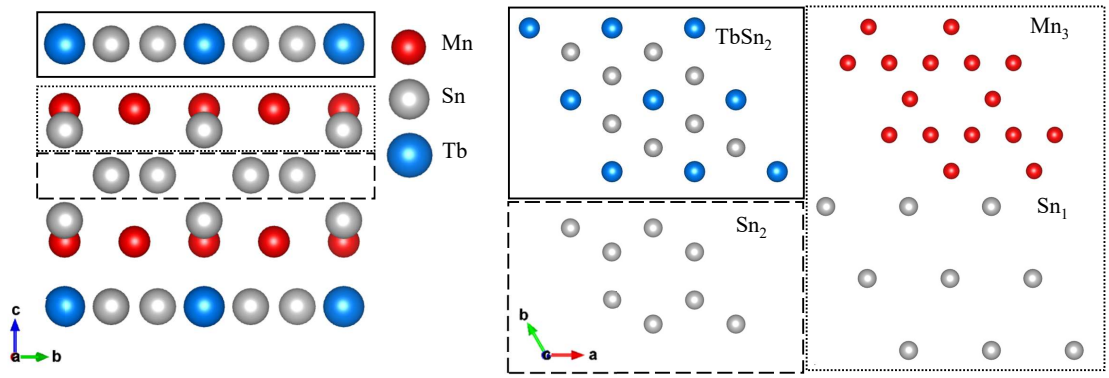


Figure 1: Crystal structure of a single unit cell of $TbMn_6Sn_6$, highlighting the Mn Kagome layer. The unit cell is composed of stacked atomic layers along the c-axis. From top to bottom, the sequence consists of a Tb–Sn layer, followed by an Mn Kagome layer coordinated with Sn, and then an Sn layer. This stacking pattern then repeats in reverse, forming a symmetric layered structure.

2.1.2 Band Structure

The investigation of the Kagome metal is driven by the need to understand the behaviour of electrons in the material, which is encoded in its electronic band structure. Band structure is most naturally analysed in momentum space rather than real space, as this representation simplifies the periodic potential of the lattice and provides clearer insight into the material's electronic properties. Momentum space is the reciprocal space of the crystal lattice, where each point corresponds to an electron wave vector $\mathbf{k}$ instead of a spatial position. The unit cell of momentum space is referred to as the Brillouin Zone. The Kagome lattice Brillouin Zone is shown in figure 2a.

To model the band structure of the Kagome lattice, a tight-binding approach is adopted, in which each atom hosts localised electron orbitals that allow hopping only to nearest neighbours, while the effects from spin-orbit coupling are ignored [4].

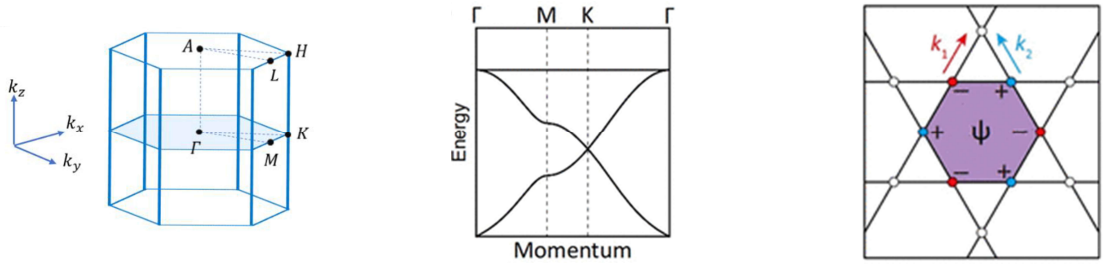
The Hamiltonian for this system has the form:

$$H(\mathbf{k}) = -2t \begin{pmatrix} 0 & \cos(\mathbf{k}\cdot\mathbf{a}) & \cos(\mathbf{k}\cdot\mathbf{b}) \\ \cos(\mathbf{k}\cdot\mathbf{a}) & 0 & \cos(\mathbf{k}\cdot\mathbf{c}) \\ \cos(\mathbf{k}\cdot\mathbf{b}) & \cos(\mathbf{k}\cdot\mathbf{c}) & 0 \end{pmatrix} \tag{1}$$

where t is the hopping parameter, $\mathbf{k}$ is crystal momentum, and $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ represent

vectors around the triangle of the Kagome lattice connecting the nearest neighbour sites [5].

Solving this Hamiltonian yields the non-trivial energy bands along high-symmetry directions visualised in figure 2b. At the K points, two of the energy bands cross linearly generating a Dirac point that gives rise to the characteristic Dirac cone structure in the energy–momentum relation. In Dirac cones, $E(\mathbf{k}) \propto \mathbf{k}$ [6], i.e. the relationship between energy and momentum is linear. Electrons near the Dirac points behave as massless Dirac fermions, enabling high-velocity charge transport with minimal scattering, which gives rise to the potential for nearly lossless conduction. The third energy band is flat throughout the entire Brillouin zone, that arises from the destructive interference of the wavefunctions from figure 2c. In flat bands, energy is constant with momentum, k and electrons have zero velocity and infinite mass, so they are effectively localised.



(a) (b) (c)

Figure 2: (a) 3D Brillouin Zone for a Kagome lattice with the high-symmetry Γ plane highlighted in blue. In this plane, the K points are at the vertices of the hexagonal zone and the M points are at the midpoints. The band structure in the plane (b) features three non-trivial energy bands.The flat band extends across all momentum space with zero gradient. Two energy bands meet linearly at the K point. (c) The wave function at the white points is a superposition of the contributions from the red and blue points. The wave function from the red point contributes negatively, and destructively interferes with the wave function from the blue point to result in a zero magnitude at the white point. Figures (b) and (c) are adapted from [4].

2.2 Photoelectron Spectroscopy

Photoelectron spectroscopy refers to experimental methods that are based on the photoelectric effect; a process by which an electron in a solid absorbs energy from an incident photon and is emitted from the solid. In this experiment, two Photoelectron Spectroscopy techniques were used: X-ray Photoelectron Spectroscopy (XPS) to understand chemical bonding environments, and Angle-Resolved Photoemission Spectroscopy (ARPES) to map the occupied electronic band structure. The process of photoemission for a given electron can be partitioned into three key steps.

2.2.1 Step 1: Photo-excitation of electrons in solid

An incident X-ray photon transfers its energy hv to electron within the solid. If the photon energy is sufficient to overcome the binding energy of the electron, E_B, then the electron is ejected from the atom as a photoelectron with kinetic energy $E_k = hv - E_B$.

2.2.2 Step 2: Propagation of the electron to the surface

The free electron can propagate in all directions through the solid, losing energy through interactions with atoms in the solid, other electrons and vibrations. The inelastic mean free path of the photoelectron relates to the kinetic energy of the electron as $\lambda \propto \sqrt{E_k}$ [7].

2.2.3 Step 3: Transmission of the electron from the surface into vacuum

The work function, Φ, is the energy needed to liberate an electron from the analyser's surface into vacuum. Each analyser has its own characteristic work function, which depends on its material, surface condition, and calibration. If the free electron reaches the surface with kinetic energy greater than the work function of the analyser, it can overcome the potential barrier and is released into free space with kinetic energy

$$E_k = hv - \Phi - E_B \tag{2}$$

and can be measured by the detector.

2.2.4 Emission angle

The emission angle θ, with respect to the z-axis, encodes information about the momenta of the electrons in the material. The momentum vector of the emitted photoelectron can be decomposed into two elements, parallel and perpendicular to the surface: $p_\parallel$ and $p_\perp$. As we are in the non-relativistic limit,

$$p = \sqrt{2m_e E_k} \tag{3}$$

where m_e is the electron mass, and so

$$p_\parallel = \sqrt{2m_e E_k}\sin\theta \tag{4}$$

From the de Broglie relationship [8]

$$\lambda = \frac{h}{p} \tag{5}$$

where λ is the wavelength of the photoelectron, h is Planck's constant and p is momentum of the photoelectron, we can relate the wave vector of the emitted photoelectron,

$$\mathbf{k} = \frac{2\pi}{\hbar}\hat{\mathbf{u}} \tag{6}$$

to the momentum of the photoelectron, as

$$\mathbf{k} = \frac{\mathbf{p}}{\hbar} \tag{7}$$

The in-plane wave-vector of the electron is hence given by [9]

$$\mathbf{k}_{\parallel} = \frac{\sqrt{2m_e E_k}}{\hbar} \sin\theta \tag{8}$$

3 Experimental

We grew thin-film samples of varying thicknesses and then analysed them at the Toroidal Analyser at the Australian Synchrotron in the research groups allocated 5-day beam time from the 13th-18th of August. The primary technique used to investigate the chemistry of the samples was X-ray Photoemission Spectroscopy (XPS). Angle-Resolved Photoemission Spectroscopy (ARPES) was also conducted to provide insight into the band structure of thin film $TbMn_6Sn_6$. The samples were transported to the facility in an ultra-high vacuum suitcase so prevent oxidisation of the sample, which is crucial for analysis.

3.1 Thin film growth

The method we use to grow samples of thin film $TbMn_6Sn_6$ is Molecular Beam Epitaxy (MBE). This cutting-edge technique allows layer-by-layer crystal growth, pivotal to understanding how the properties of $TbMn_6Sn_6$ change with thickness. The MBE chamber is kept in an ultra-high vacuum (10^{-9} mbar) environment with a series of pumps. Crucibles containing the elemental sources are heated to their sublimation temperatures, generating molecular beams of our base elements which are directed at a heated Silicon substrate prepared with a 7x7 surface reconstruction. Upon collision, the atoms bind to the Silicon surface and the Mn sub-lattice forms the Kagome network within the crystal structure. This process allows epitaxial synthesis of the Kagome lattice, with minimal impurities from the environment.

3.1.1 Silicon surface reconstruction

The silicon 7x7 crystal structure is created through flash annealing. This highly ordered structure is the lowest energy configuration of the Si (111) surface. Direct current heating through the silicon wafer allows the atoms to move, and the atoms organise into the 7x7 structure as the current is cut and the silicon cools quickly. The established flash annealing process is as follows:

1. 0.0A to 1.6A in quick 0.2A steps
2. At 1.6A, maintain until stable temperature reached
3. 1.6A to 2.0A in quick 0.2A steps

4. At 2.0A, maintain for 10s
5. 2.0A to 6.0A in quick 1.0A steps
6. At 6.0A, maintain for 20s
7. 6.0A to 2.5A in 30s 0.5A steps
8. 2.5A to 1.6A in quick 0.1A
9. At 0.1A, maintain for 2min

The flash annealing process is repeated multiple times to ensure an atomically flat substrate with large terraces.

3.1.2 Growth method

After the Silicon substrate has been prepared with a 7x7 surface reconstruction, it is heated to the growth temperature. This surface reconstruction of silicon is highly reactive, so a buffer layer of Mn3Sn is grown first, to a thickness of 1.5 nm. Then, the deposition of atoms is controlled according to a growth recipe.

The flux rate for the molecular beams is calibrated by releasing the molecular beam onto a quartz crystal microbalance. The crystal has a specific resonant frequency when a current is applied, and measuring the change in this frequency gives information on the deposition rate of the molecular beam.

Considering growth rates, the growth recipe is developed to approximately reflect the desired stoichiometric ratio for each layer of the $TbMn_6Sn_6$ structure. There are other considerations for the growth recipe. Firstly, optimising the substrate temperature to ensure that incident atoms have the mobility to settle into the correct lattice structure without de-sorbing from the surface due to excessive kinetic energy or forming unwanted crystal structures. Also, the pressure in the MBE chamber must be restricted, whilst maintaining the high sublimation temperatures for the base elements increases the temperature of the chamber, and the pressure with it.

The growth recipe used for sample preparation was Tb@ 2.9A/min, Mn@ 3.3A/min, Sn@ 1.1A/min.

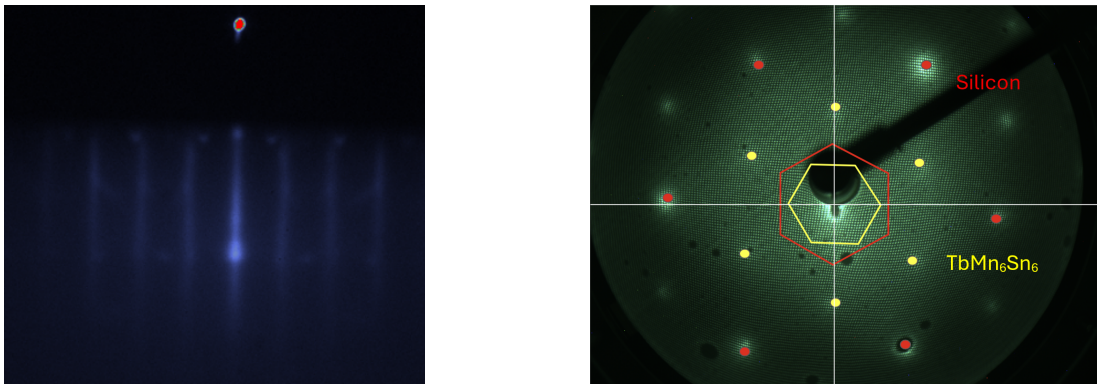
3.1.3 Reflection High Energy Diffraction

During growth, the quality of the sample is measured with in situ Reflection High-Energy Diffraction (RHEED). An electron gun is directed at a grazing angle to the film and incident electrons diffract from the surface atoms, interfering to create a diffraction pattern on the detector screen. High-quality layered films produce vertical streaks on the detector, with three-dimensional islanding appearing as spots. Islanding impacts the quality of the sample and hence it is important to minimise. Imperfections in the silicon 7x7 crystal

structure, vacuum levels and substrate temperature all contribute to islanding and need to be controlled. Thicker samples have less islanding, as the atoms have more time and energy to move around and fill the gaps so layers even out over time. The RHEED diffraction pattern of the 2-unit cell sample used in analysis is shown in figure 3a, which was used assess that the growth was high-quality.

3.1.4 Low-Energy Electron Diffraction

Low-Energy Electron Diffraction (LEED) was also used to study the crystal structure of the samples. Like RHEED, this technique uses an electron beam to create a diffraction pattern and hence show the crystal structure as seen in figure 3b. In LEED analysis, the electron beam is perpendicular to the surface of the sample. The diffraction pattern allows for calibration of the orientation of the sample for further analysis techniques at the Toroidal analyser.



(a) (b)

Figure 3: (a) RHEED pattern of the 2-unit cell sample. The prevalence of streaks over dots indicated a high-quality sample. (b) LEED pattern of the 6-unit cell sample with the diffraction patterns from the Silicon 7x7 surface shown in yellow, and those from the $TbMn_6Sn_6$ sample in red.

3.2 X-ray Photoemission Spectroscopy

X-ray Photoelectron Spectroscopy (XPS) is a well-established and widely used technique for analysing the chemical composition of materials. It operates by irradiating a sample with X-rays to excite core-level electrons in the atoms. When these electrons are ejected from the material, their kinetic energy is measured and their binding energy is determined via equation (2). XPS spectra of photoelectron intensity against binding energy can hence be produced.

Each element has characteristic binding energies for its core-level electrons, and these values are sensitive to the chemical environment. Therefore, XPS spectra encode the elemental composition and chemical state of atoms. Electrons that undergo inelastic scattering before reaching the surface of the material will not have this characteristic binding energy and hence do not contribute to the XPS peak for the given core level [10].

Electrons are increasingly likely to scatter as the depth of the material increases, hence XPS spectra only include the surface features.

In addition to qualitative analysis, the stoichiometry of a compound can be extracted by integrating the area under the spectral peaks and applying relative sensitivity factors.

3.2.1 Peak fitting

XPS spectra are produced from the data and analysed using IGOR software . Before peak fitting and understanding of chemical bonding environments can take place, background subtraction is performed. Inelastic scattering of electrons results in non-linear background for XPS, so the Shirley method [11] of background subtraction is applied using the inbuilt IGOR fit assistant. The peaks are then fitted with a Voigt profile, referencing existing literature to determine their positions. Spin orbit coupling means that the peaks for these core levels are doublets with two possible states corresponding to the value of the spin angular momentum quantum number, s, being $\pm 1/2$. The area ratios of the primary and secondary peaks are given by the ratio of the degeneracy, g, of these states, where $g = 2j + 1$. The total angular momentum number, j, is given by $j = l + s$ where l is the angular momentum quantum number. For example, the core level 4d for Tb was analysed, and it has $l = 2$, $j = 3/2, 5/2$ and hence a doublet ratio of $2 : 3$. Core levels 3p and 3d analysed have doublet ratios of $1 : 2$ and $3 : 4$ respectively. The doublet shifts are estimated using previous literature values.

3.2.2 Peak areas

The area under an XPS spectral peak is directly proportional to the number of atoms of a given element present near the surface of the material. By comparing the areas of peaks corresponding to different elements, it is possible to determine the stoichiometric ratios in the sample. However, since different electron orbitals and elements have different probabilities of photoemission, the raw peak areas must first be normalised using the corresponding photo-ionisation cross-section, $\sigma(hv)$,This parameter represents the effective area presented by an atom for a photon with energy hv causing ionisation of an electron from the orbital it concerns. The normalised peak area is hence given by

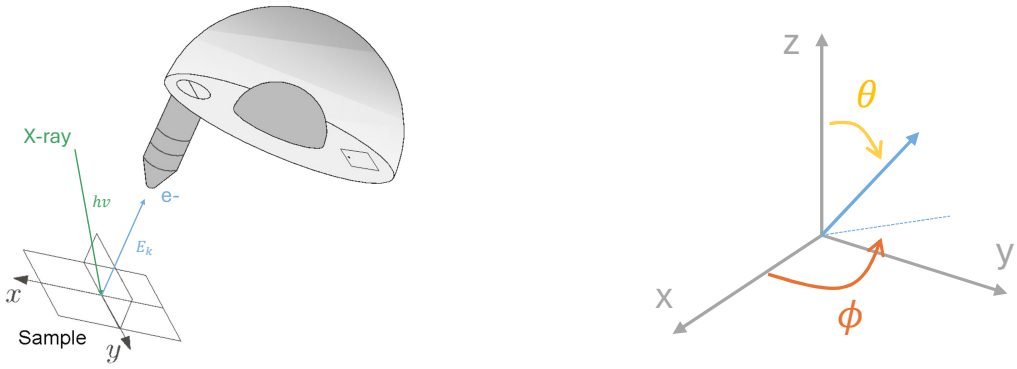
$$A = \frac{A_0}{\sigma(hv)} \tag{9}$$

where A_0 is the original peak area.

3.3 Angle-Resolved Photoemission Spectroscopy

Angle-Resolved Photoemission Spectroscopy (ARPES) is a powerful Photoemission Spectroscopy technique for studying electronic band structure. Monochromatic light with photon energy, hv, greater than the work function, Φ, is focused on the surface causing photoelectrons to be emitted with energy E_k and emission angle with respect to the z-axis,

θ. The detector of the ARPES instrument used in this experiment has a full hemisphere, as seen in figure 4, so emission angles ϕ and θ can be resolved. The kinetic energy is transformed to binding energy from energy conservation as equation 2 and the in-plane wave-vector of the electron is hence given by equation 8. After these transformations, it is possible to produce an intensity map of binding energy as a function of wave vector $k_{\|}$. The band structure of the material is directly encoded in this map, and analysis of features near the Fermi energy reveals the predicted energy bands.



(a) (b)

Figure 4: (a) Schematic of the ARPES setup. Incident X-ray photons with energy hv eject electrons with kinetic energy E_k. (b) The photoelectron is emitted at a polar angle θ relative to the z-axis and an azimuthal angle ϕ relative to the x-axis.

Like XPS, ARPES is inherently surface-sensitive, probing just a few atomic layers due to the short mean-free path of electrons at relevant photon energies. While this can be a limitation in bulk systems, it becomes a key advantage in the study of thin films, where the relevant electronic properties are near the surface.

Synchrotron-based ARPES has the advantage of variable photon-energy, which makes it possible to map the electronic structure across the entire 3D momentum space. Variation of the incident photon energy allows for different sections of the 3D Brillouin zone to be analysed. The photon energy is related to the z momentum, k_z, as

$$hv = \frac{\hbar^2 k_z^2}{2m} + \Phi - V_0 \tag{10}$$

where Φ is the work function and V_0 is the inner potential for the material. For $TbMn_6Sn_6$, $V_0 = 14.4\ eV$For the ARPES instrument used, the resulting k_z - hv relationship is plotted in figure 5.The Γ and A planes are of interest due to their high symmetry.

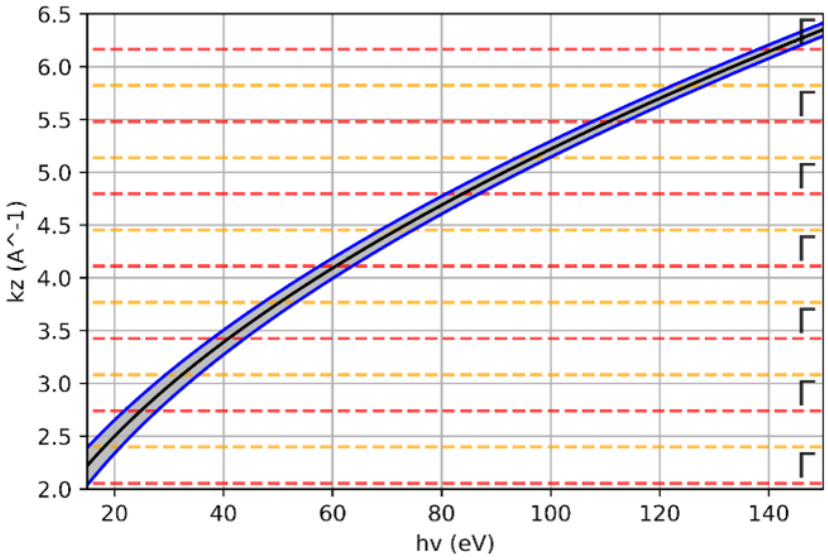


Figure 5: Theoretical predictions for k_z values against photon energy, hv, for $TbMn_6Sn_6$ with error of 1 standard deviation is blue. The k_z values corresponding to the high-symmetry planes are marked in yellow and red dotted lines for the A and Γ planes, respectively.

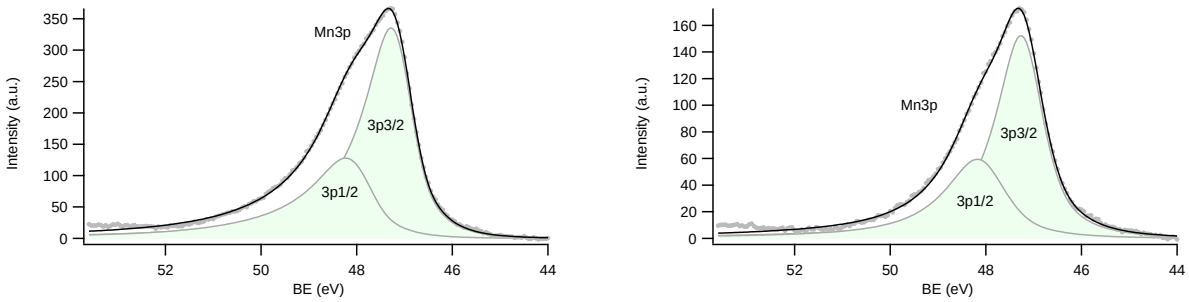
4 Results and Analysis

4.1 X-ray Photoemission Spectroscopy Analysis

A photon energy of 160 eV was used to measure the core levels Mn 3p, Sn 4d and Tb 4f and produced XPS spectra by integrating the intensities over an angle of emission window of $-60° < \theta < 60°$. These core levels are chosen for analysis because of their low binding energy range corresponding to a low incident photon energy range that can be captured in this toroidal analyser system. XPS spectra of a 16-unit cell sample at this photon energy was also taken as a reference.

4.1.1 Mn 3p Core Level

The spectra for the Mn 3p core level 2-unit cell sample showed a clear doublet, which is fit with a doublet Voigt profile in figure 6. The fitted peak position for the $3p_{3/2}$ peak is 47.32 eV, in line with the binding energy range from the literature of $47.0 - 49.4$ eV [12, 13]. The doublet separation is 0.94 eV with an expected doublet ratio of 1 : 2. The XPS data for the 6-unit cell sample was fit with the same method in line with the expected literature values.

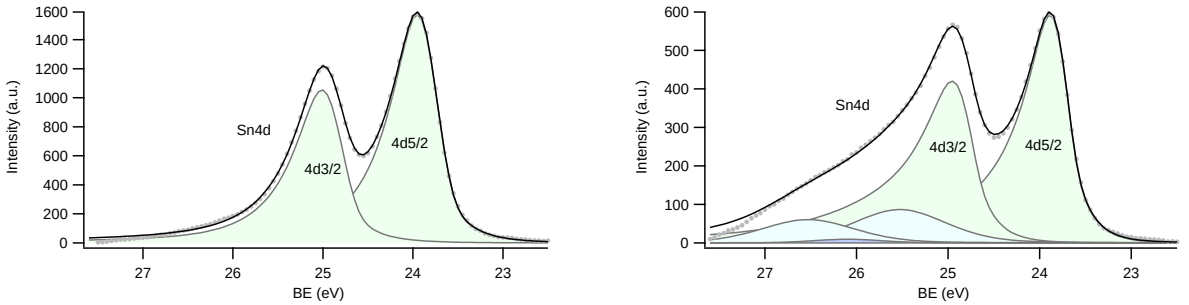


(a) 2-unit cell sample (b) 6-unit cell sample

Figure 6: Mn 3p core levels for the (a) 2-unit cell and (b) 6-unit cell samples with the individual doublets in green and the total fit as a solid black line. The data is in grey.

4.2 Sn 4d Core Level

The Sn 4d core level spectra for the 2- and 6-unit cell samples differ significantly. The data for the 2-unit cell sample indicated a clear doublet peak, which is successfully fit with a single doublet in figure 7a. The fitted peak position is 24.00 eV with a doublet separation of 1.06 eV and a doublet ratio of 2 : 3. The general shape of the data and these values are in line with expectations for Sn 4d [14, 15, 16]. In contrast, the 6-unit cell sample exhibits and extra feature, a shoulder at increased binding energy.. The shape resembles spectra produced by Padova *et al.* [17] for oxidised Sn, which was fitted with three doublet peaks corresponding to Sn, Sn^{2+} and Sn^{4+}. Taking the separation values of these oxide peaks from the primary peak, the spectrum was successfully fit. The peaks for the cations have larger binding energies, reflecting the increased nuclear charge experienced by the core electrons.



(a) 2-unit cell sample (b) 6-unit cell sample

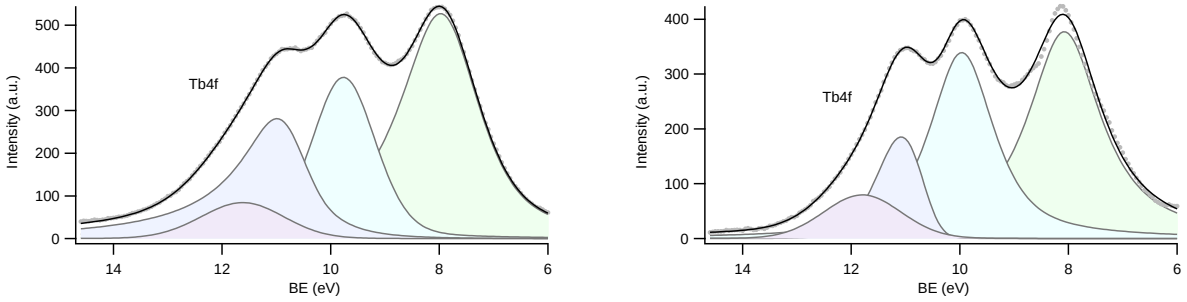
Figure 7: Sn 4d core levels for the (a) 2-unit cell and (b) 6-unit cell samples for the XPS data in grey. The doublet for the Sn peak is in green, with the doublet corresponding to a Sn^{2+} environment in blue and the doublet corresponding to the Sn^{4+} environment in purple. The Sn^{4+} peak is small as compared to the other peaks, so is barely visible. The total peak fitting is a solid black.

4.2.1 Tb 4f Core Level

The fitted Tb 4f core levels for both sample thicknesses are displayed in figure 8. As with the Mn 3p and Sn 4d core levels, a single doublet is expected from the $4f_{5/2}$ and

$4f_{7/2}$ orbitals from spin splitting, especially given Tb has only one bonding environment in the lattice, Tb-Sn, however the spectra have at least three distinct peaks. A thorough literature review was conducted on the Tb 4f environment, and the studies are extremely limited. The NIST X-ray Photoelectron Spectroscopy Database is an extensive database of published literature values for peak positions, and has no entries for Tb [18].

Additionally, XPS analysis for the bulk crystal $TbMn_6Sn_6$ investigated a different Tb core level. Following the fitting method demonstrated in the XPS Handbook by XPS International LLC [19], I successfully fit the spectra with 4 singlet peaks confirming that the strange features are expected. The resource gave no motivation for this fitting method, and it does not have a physical basis, given we expect doublet peaks in XPS spectra. The Tb 4f core level is at a very low binding energy, close to the valence bands, so the peak is likely be contributed to by more than the Tb 4f orbital, i.e. deep binding energy valence bands. As a result, taking the peak area of this orbital may lead to an overestimation of the abundance of Tb in the sample.



(a) 2-unit cell sample (b) 6-unit cell sample

Figure 8: Tb 4f core levels for the (a) 2-unit cell and (b) 6-unit cell samples fit with four singlet peaks against the data in grey.

4.2.2 Origin of additional peaks in the 6-unit cell sample

Although the methods of Sn 4d core level fitting used by Padova *et al.* [17] for oxidised samples provided a successful fit, it is unlikely that the 6-unit cell sample was oxidised. Like the 2-unit cell sample, it was prepared, transported, and measured under ultra-high vacuum conditions. Moreover, if oxidation had occurred, additional features would be expected in the Mn 3p spectrum, which are not observed. Finally, the Sn 4d XPS spectra for the 16-unit cell sample also had a slight shoulder, less extreme than that of the 6-unit cell sample, but still indicative of an alternate bonding environment.

Another consideration for analysing the Sn 4d level is the bonding environments of Sn. Sn has three distinct bonding environments in the $TbMn_6Sn_6$ lattice. The typical core level peak corresponds to an Sn-Sn bonding environment however Sn also has Sn-Tb and an Sn-Mn metallic bonds.

When an element is in a bonding environment with a different element, differences in electronegativity can affect the XPS spectra. The more electronegative element will attract more of the electron cloud, leading to greater shielding of the nucleus. This makes it easier to liberate electrons, leading to a shift to a lower binding energy. The electronegativity

value for Sn on the Pauling scale is 1.96, larger than Tb, 1.1 and Mn, 1.55 [20]. Hence, if the Sn XPS spectrum is affected by its bonding environment, a shift to a lower binding energy would be expected, not the shift to higher binding energy that is present.

4.2.3 Peak Area Ratios

To determine the stoichiometric ratio of the samples, peak areas were first normalised with the corresponding cross-section values [21, 22] before the ratio was calculated. The Mn 3p peaks from figure 6 were used with $\sigma(160eV)_{Mn3p} = 1.031$. The peak from figure 7a and the sum of the peaks from figure 7b were used for Sn 4d with $\sigma(160eV)_{Sn4d} = 0.460$. The Tb 4f core level was fit with a single doublet peak corresponding to the expected literature value for the binding energy of $7.5eV$ [19], with the extra features encompassed by a single peak. The peak area for the doublet peak was used as an improved estimation for the peak area of this environment, normalised with $\sigma(160eV)_{Tb4f} = 5.338$.

The ratios calculated can be seen in table 1. The ratios deviate significantly from the expected ratio of 1:6:6. The thinner, 2-unit cell film shows an increase in Sn:Mn ratio. This indicates a higher Sn concentration near the surface of the films. This could suggest that the growth process may favour Sn-rich surface layers. Also, since XPS is a surface-sensitive technique, this could indicate that the surface termination is an Sn layer. When comparing the Tb ratio of the two two films, the 6-unit-cell sample showed slightly stronger Tb and Sn peaks relative to Mn, meaning it more closely approaches the bulk stoichiometry as the film becomes thicker. This trend is consistent with a gradual transition from surface-dominated composition in the thinner film to more bulk-like behaviour in the thicker film.

Studying a cleaved bulk crystal of $TbMn_6Sn_6$ with a known stoichiometric ratio could be used a baseline for comparing the quality of our growths.

Sample	Tb	Sn	Mn	Sn:Mn
2-unit cell	1.00	14.84	0.86	17.3
6-unit cell	1.00	8.03	3.12	2.5

Table 1: Calculated peak area ratios for the 2- and 6-unit cell samples. Includes Sn:Mn ratio.

4.3 Angle-Resolved Photoemission Spectroscopy Analysis

Key features of the predicted Kagome band structure are observed in the samples, as shown in figure 4.3c, d, at photon energies of 87 eV and 89 eV.

At 87 eV, a nearly non-dispersive band is visible across the sampled in-plane momentum range. This broad, flat feature is consistent with the flat band predicted in Kagome systems, where electronic states become highly localised due to destructive interference inherent to the lattice geometry. The width of the flat band is < 200 meV, consistent with prior reports on rare-earth RMn_6Sn_6 Kagome metals [23]. The flat band appears

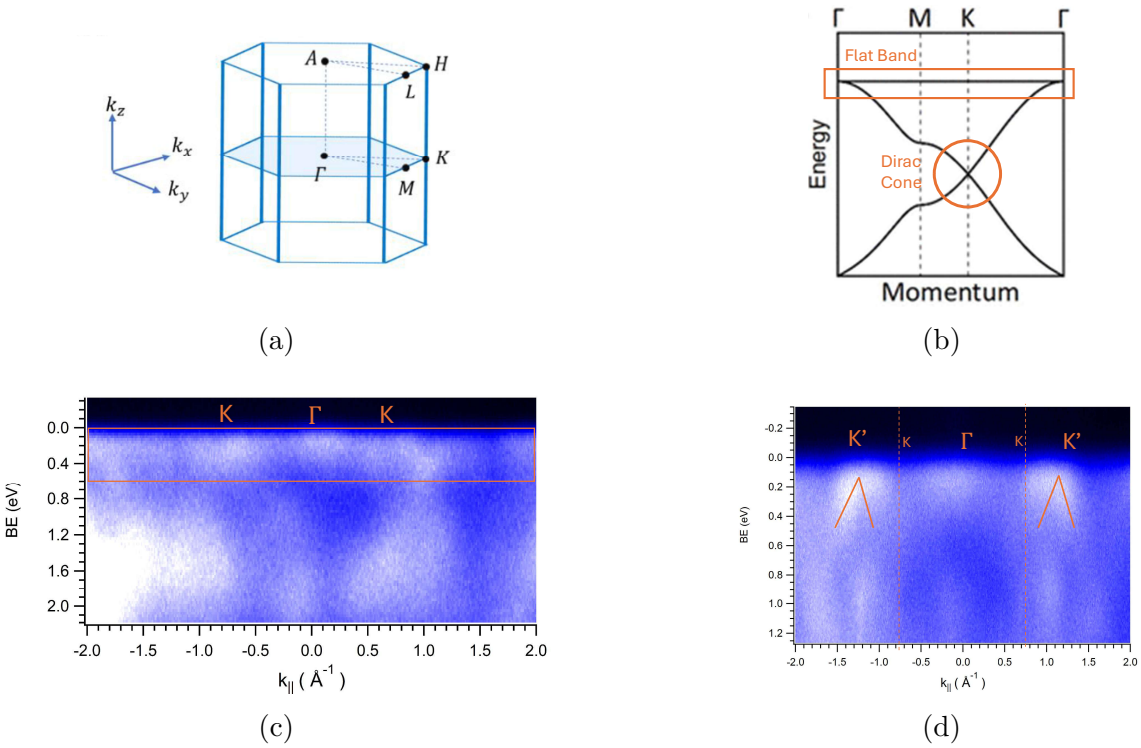


Figure 9: (a) 3D Brillouin Zone of $TbMn_6Sn_6$ with momentum space axes and the high-symmetry Γ plane highlighted in blue. (b) The band structure for this Γ plane with the flat band and the Dirac point circled in orange and labelled. ARPES spectra of the (c) 6-unit cell sample at $87eV$ corresponding to the $K - \Gamma$ cut featuring a flat band and (d) the 6-unit cell sample $89eV$ featuring Dirac cones at the second K points.

to be located at about 200 meV below fermi energy, close to reported values for bulk RMn_6Sn_6.

At 89 eV, the high-intensity regions near the Fermi energy form shapes resembling Dirac cones, particularly at a parallel momentum value of approximately 1.5 Å^{-1}, which corresponds to the second K-point in the extended Brillouin zone, as opposed to the first K-point, located at 0.76 Å^{-1}.

Theoretical calculations from figure 5 predict that the Γ-plane is accessed at photon energies around 88 eV, placing these measurements within the correct k_z slice to reveal Kagome-specific band structure features. The simultaneous observation of both flat bands and Dirac cones within this photon energy range strongly supports the presence of a well-formed Kagome electronic structure in the sample.

5 Conclusions and Future Plans

XPS and ARPES were used to study the chemical stoichiometry and electronic band structure of thin-film $TbMn_6Sn_6$. The samples showed an increased Sn:Mn ratio from the 6- to the 2-unit cell sample indicating a Sn termination layer or a Sn-favouring growth method. The interesting features of a Dirac cone and a flat band were identified in the

band structure of the material.

Future plans include further XPS analysis to determine the origin of the extra feature in the Sn 4d core level for the 6- and 16-unit cell samples. Firstly, by comparison of XPS spectra at different Sn core levels, it will be able to be determined if there is a consistent feature shifted to higher binding energies. This would suggest the feature is a consequence of the bonding environment of Sn. The Sn 3d core level at 484.9 eV [19] is particularly suitable for this analysis due to its strong, well-separated peaks. Literature reports a peak separation of 8.4 eV [19] for this transition, which minimises peak overlap and should make secondary features easier to identify. Additionally, no Tb or Mn core levels lie near this energy, reducing the likelihood of spectral interference. Secondly, depth-dependent XPS, by which different depths of the sample are probed by varying the photon energy used in analysis, will establish if the feature is just a surface-feature or a bulk-feature. The photon energy of 160 eV used in this experiment probes core to the surface. Thirdly, XPS analysis on the bulk crystal at the same photon energy for the same core levels would serve as a good point of comparison for this core level, and also for analysis the Tb 4d core level.

Accurate spectral fitting is essential for reliable determination of elemental ratios. XPS measurements of pure Tb metal would enable more precise fitting of the Tb 4f environment. From here, it would be determined which features arise from the unique bonding environment of the lattice, assisting with identifying the chemical environment.

Higher-resolution ARPES scans to be taken at the Shanghai Synchrotron Radiation Facility will be taken at the identified photon energies of interest. These will confirm the electronic band structure features that were predicted and tentatively observed in the current dataset.

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